AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 30 · STUDENT LESSON

Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Specify inputs. Trace the baseline. Compare under one contract.

Continue the activity project with a complete catalogue, explicit fitting rules and distinct no-match and clarification paths.

Keep one habit: Validate → check every condition → choose in order → keep the evidence.

MAIN LESSON · SECTION 1

What exactly must go in—and come out?

> Turn the observed need into a repeatable input/output contract.

Start with an agreement test

Two people should be able to read the same request and agree on the required result. Specify valid inputs, every condition, the tie rule and the difference between invalid input and a genuine no match.

Recall why zero wrong suggestions is insufficient

A recommender always reports no match. Why is a count of zero in**compatible** suggestions insufficient to show that it helps users?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can turn the project outcome into precise **input** and **output requirement**s.
- I can apply a simple manual **baseline** to valid requests and record its results.
- I can plan a fair comparison using the same catalogue, requests and judging rules.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

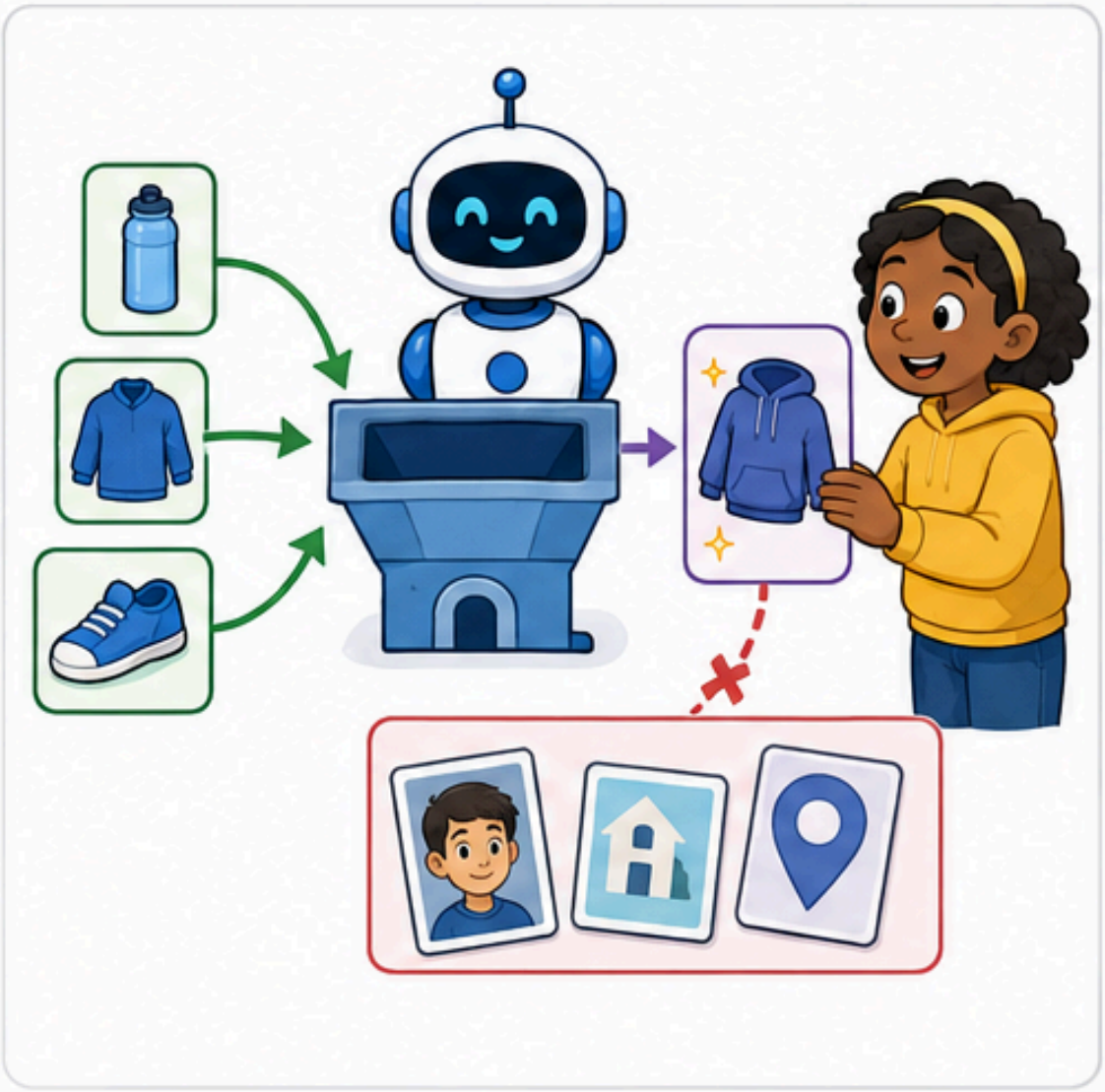
Remember before reading

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

A useful design makes required information and promised result explicit.



LEARNING TARGET Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can specify valid **input**s, exact **output**s and a manual **baseline**.

MAIN LESSON · SECTION 2

Specify the activity recommender’s contract

> Continue the fictional activity project from Week 29.

Turn a vague outcome into requirements

Last week we identified a difficulty: some fictional activity choices did not fit the available time or room conditions. Now define the task precisely enough that two reviewers can agree whether an **output** is correct.

A **requirement** such as “suggest a good activity” is too vague. State which **input**s are accepted, what counts as **compatible** and what to do when no option fits.

Define valid minutes and the quiet requirement

For this paper project, a valid request contains an available time from 0 to 30 whole minutes and a quiet-room **requirement** written Yes or No. Yes means the activity must be marked quiet.

No means either quiet or non-quiet activities are permitted; it does not mean noise is required. A missing or invalid value needs clarification, not a guessed value or a no-match claim.

Define compatibility, order and output limits

A **compatible** activity takes no more than the available minutes and meets the quiet-room condition. If several activities fit, suggest the first in catalogue order. If none fits a valid request, report no match.

Give a brief reason. This is a suggestion only: it does not book resources, tell anyone to begin or guarantee enjoyment.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Read the complete approved activity catalogue

> These B1–B4 activity codes belong to this catalogue, not the later storage-bin example.

The four-row activity catalogue

Catalogue order and activity	Minutes required	Quiet activity
B1 Bookmark drawing	10	Yes
B2 Story circle	8	No
B3 Pattern puzzle	12	Yes
B4 Tower challenge	15	No

State the paper model and tie rule

These four rows are the complete approved catalogue for the paper model. Treat their times as fixed task values, not measured guarantees about real pupils. All required materials are assumed available.

The catalogue order B1, B2, B3, B4 is the **tie rule**; it is not a claim that B1 is everyone's favourite.

Trace three valid requests

For 10 minutes with quiet required, B1 fits because 10 is no more than 10 and B1 is quiet. B2 fails the quiet **requirement**. B3 and B4 take too long. For 8 minutes with quiet not required, B2 fits; B1 takes too long.

A request for 7 minutes has no **compatible** activity in this catalogue, even when quiet is not required.

Check equality, both conditions and missing values

- Check each condition. A fast activity can still fail the quiet **requirement**. A quiet activity can still take too long. Exact equality is allowed: an activity listed as 10 minutes fits 10 available minutes.

  A blank time is different from a valid time of zero.

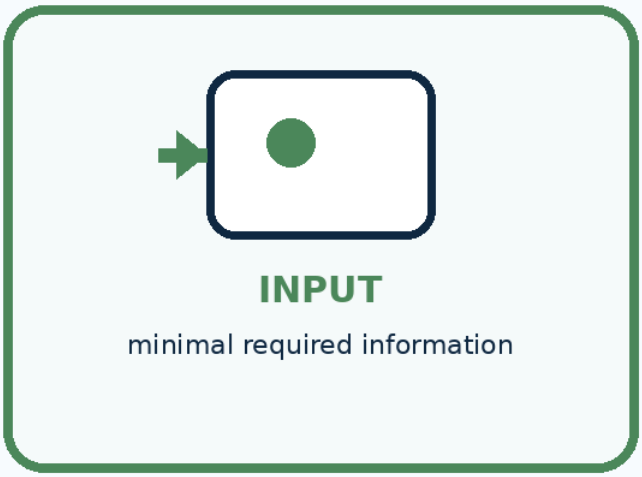
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

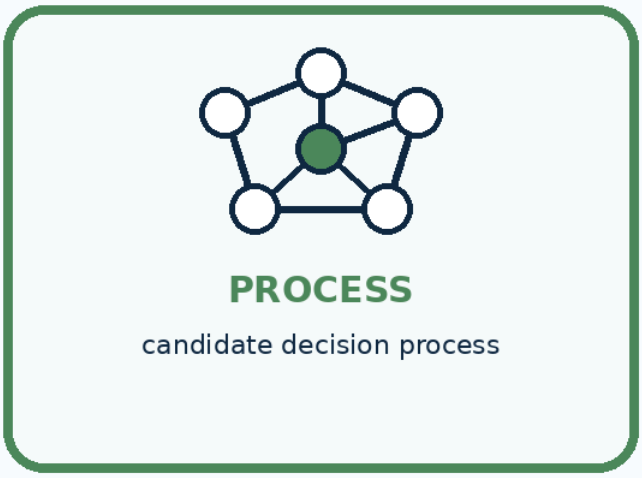
▼ Earlier wording, alternatives and picture transcripts

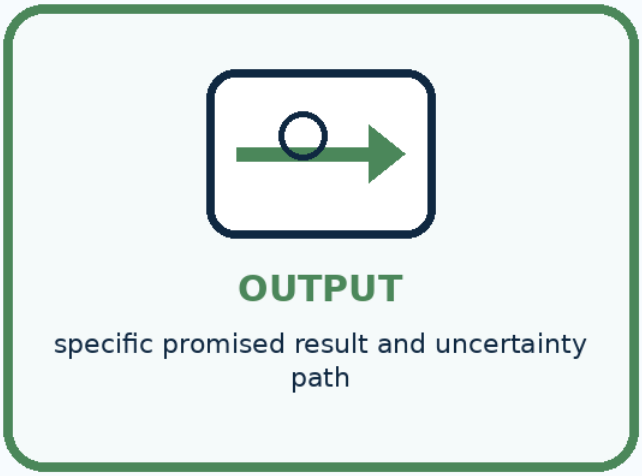
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT minimal required information

PROCESS candidate decision **process**

OUTPUT specific promised result and uncertainty path

CORE IDEA A useful design makes required information and promised result explicit.

UNDER THE HOOD Interface contracts define **input**s, **output**s, types, allowed values, **missing-data behavior**, and guarantees.

Good design avoids collecting data that cannot improve the promised result and avoids promising outcomes the **process** cannot support.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare useful fields with unsupported promises

> Collect only inputs that support the actual output contract.

Pictures paired with their evidence



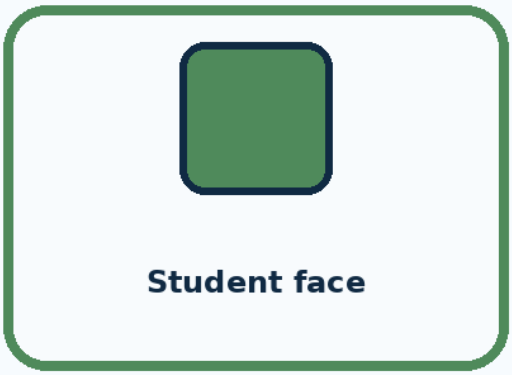
The case to investigate

CASE 1 — Item type Clue: directly supports bin choice. Meaning: required candidate.



The case to investigate

CASE 2 — Location found Clue: may disambiguate. Meaning: optional candidate.



The case to investigate

CASE 3 — Student face Clue: not needed. Meaning: remove.



The case to investigate

CASE 4 — Guaranteed owner Clue: unsupported promise. Meaning: revise **output**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Every collected field creates cost and risk; every promised **output** needs **evidence**.

▼ Earlier wording, alternatives and picture transcripts

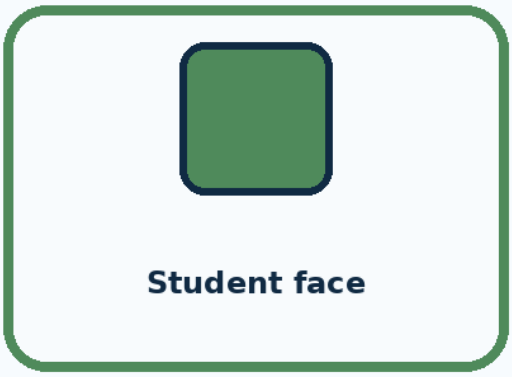
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Item type Clue: directly supports bin choice. Meaning: required candidate.

CASE 2 — Location found Clue: may disambiguate. Meaning: optional candidate.

CASE 3 — Student face Clue: not needed. Meaning: remove.

CASE 4 — Guaranteed owner Clue: unsupported promise. Meaning: revise **output**.

Change one thing in the pictured case

Change: Change request 6 to 5.

Prediction: The **baseline** returns B1.

Reason: Its capacity equals the request, and at least includes equality.

Trace the separate two-bin baseline

> Here B1 and B2 are bins with capacities 5 and 9, not the activity catalogue's codes.

Read the bin catalogue and input contract

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

Return to the activity baseline's four-stage checklist

AI ACADEMY / AI-1/30 · 03 · INSPECT THE LESSON MODEL

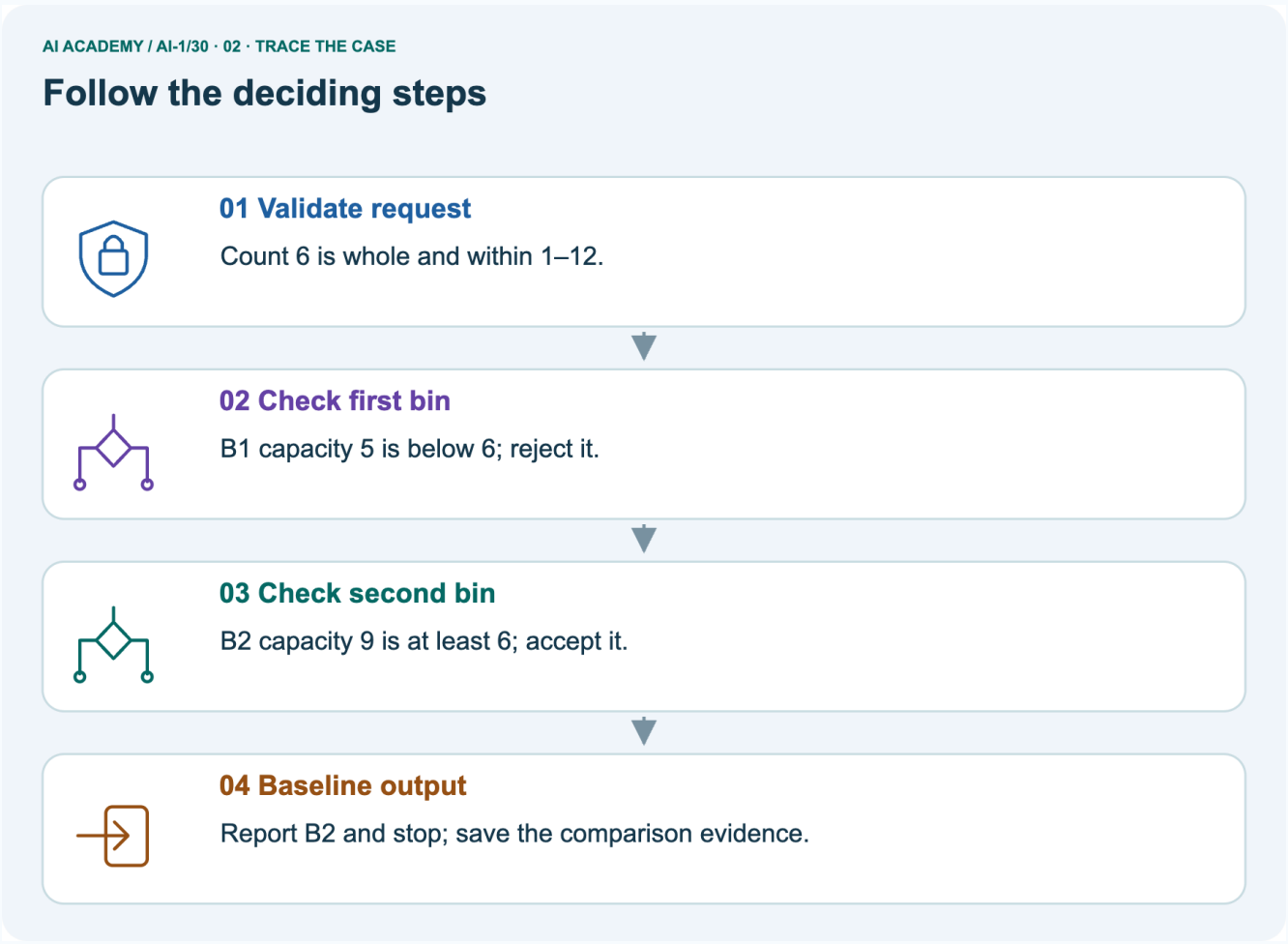
The manual checklist baseline

Stage	Manual action	Recorded result
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the six-part bin request



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate request	Count 6 is whole and within 1–12.
2. Check first bin	B1 capacity 5 is below 6; reject it.
3. Check second bin	B2 capacity 9 is at least 6; accept it.
4. Baseline output	Report B2 and stop; save the comparison evidence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Distinguish valid no-match from invalid input

- The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

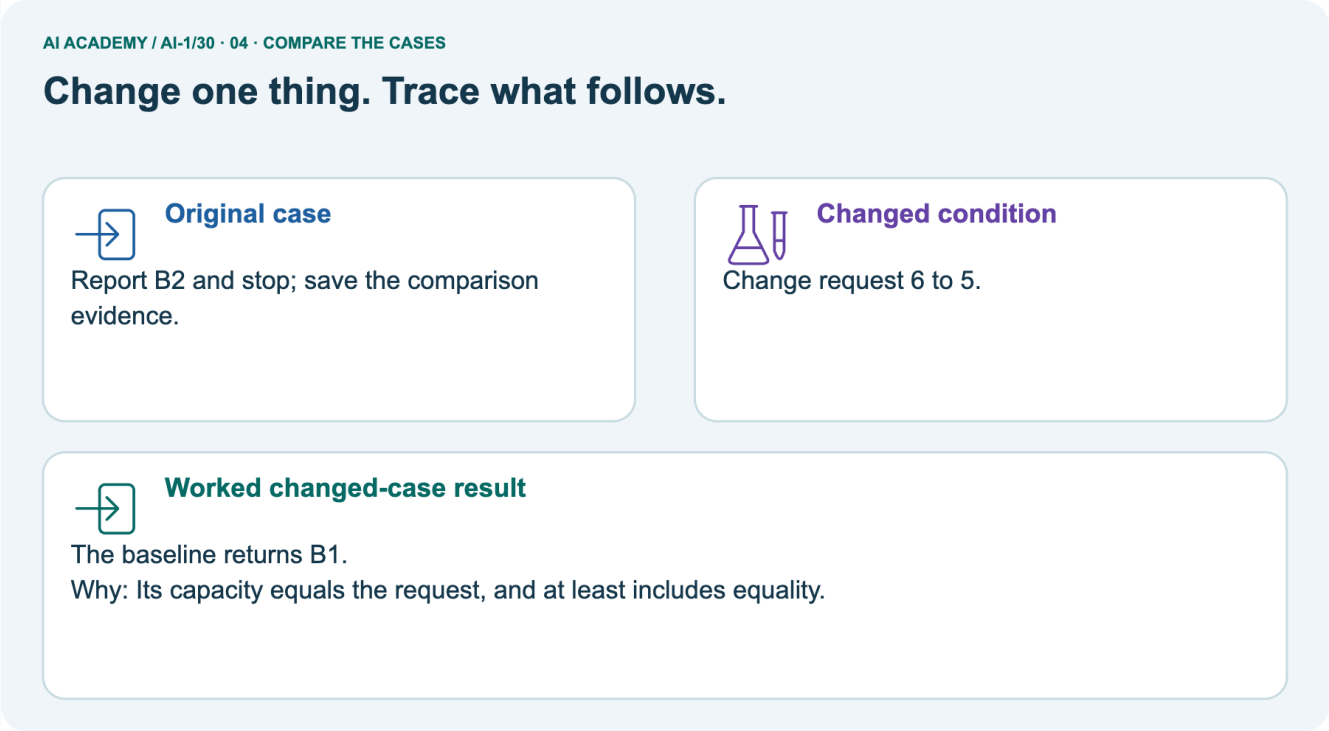
  Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

- Read the labelled picture: Trace requests 10 and blank. Why should they not receive the same explanation?

Predict the five-part boundary

- Before opening the changed-count picture, change the storage-bin request from 6 parts to 5. Which first bin has enough capacity? Explain why equality is accepted and why the activity catalogue must not be used for this separate example.

▼ Check the changed request



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the promised fitting criterion

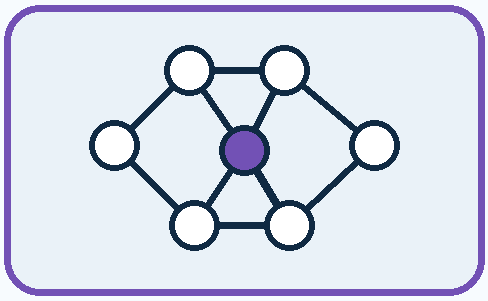
Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

STEP 1 — OBSERVE minimal required information

STEP 2 — TRACE candidate decision **process**

STEP 3 — CONCLUDE specific promised result and uncertainty path

MODEL REASONING Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

DO NOT OVERCLAIM Every collected field creates cost and risk; every promised **output** needs **evidence**.

What new **evidence** could change this answer?

A closer explanation

The manual checklist baseline

STAGE	MANUAL ACTION	RECORDED RESULT
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Write the request at the top of a sheet. If it is invalid, record what needs clarification and stop. Otherwise mark each catalogue row as **compatible** or in**compatible** using both conditions.

Circle the first **compatible** row in order, or write no match if every row fails. Record the **output** and its reason. This checklist is our **baseline**.

The **baseline** can be correct on all cases when applied carefully. A prototype that matches it has not demonstrated better compatibility.

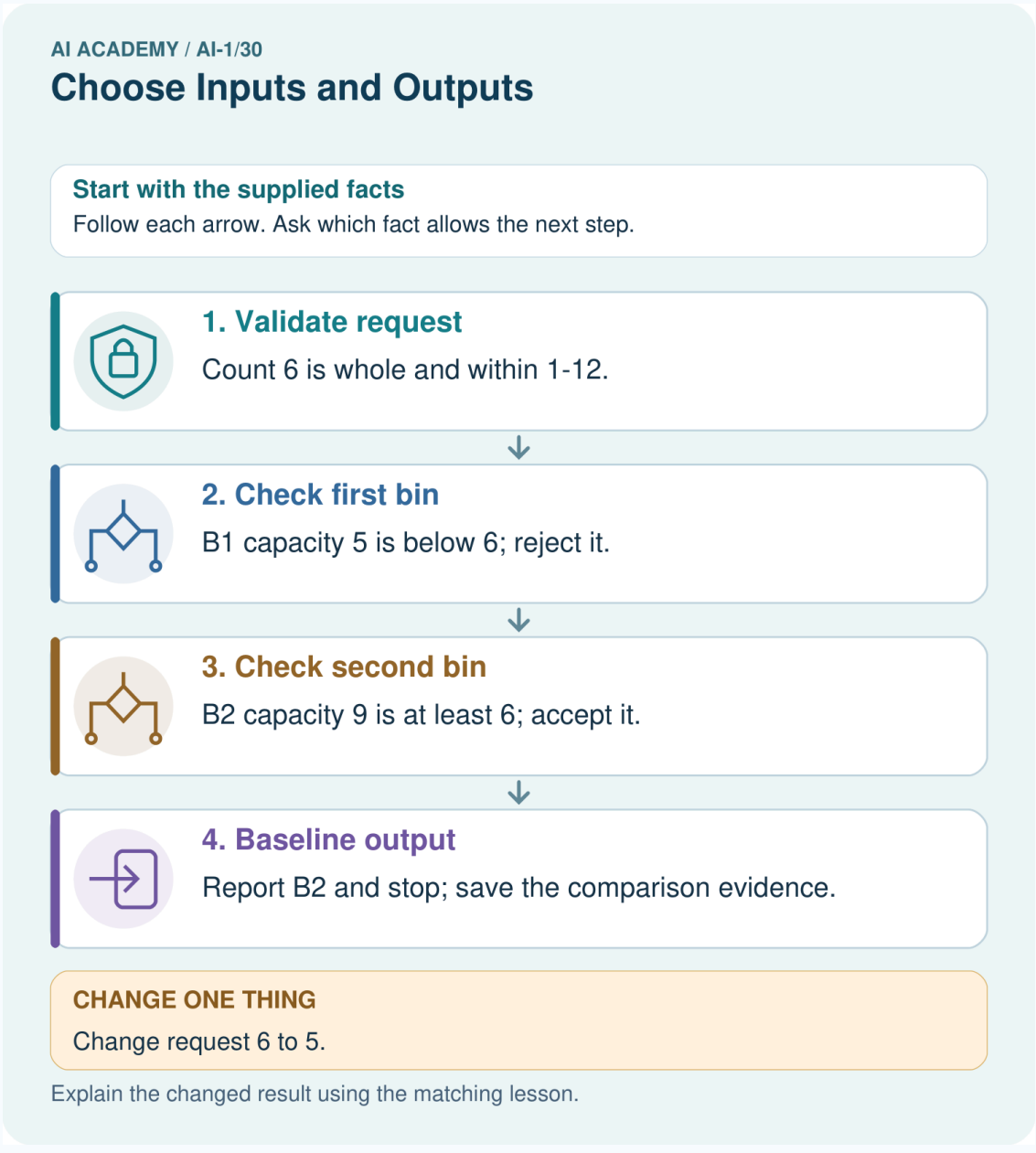
It may still be easier to use or faster to trace, but those are separate claims that need their own measures. Do not make the **baseline** skip a condition just to make the prototype look better.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Choose Inputs and Outputs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Change request 6 to 5. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Keep a repeatable baseline record

> Record the actual input, output, reason, catalogue and method version.

Build the project evidence record

Your project record now needs the problem brief, this catalogue, the **input** rules, compatibility rule, **tie rule** and clarification behaviour.

Add a **baseline** sheet with one row per request: supplied **input**, actual **output**, reason and whether the **requirement**s were met. Record the method and catalogue version so a later reader can repeat the comparison.

Keep this week's baseline separate from next week's process design

For this week, complete the **baseline** before building a new **process**. Next week you will design the full decision path, including invalid **input** and no-match cases. Keep compatibility claims within the paper model.

Real materials, changing room conditions and pupils' preferences would require **more information** and a revised task.

Repair both tempting claims

- CLAIM A More **input**s make every design more accurate.
- CLAIM B An **output** label can promise more than the **process** actually knows.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Item type	directly supports bin choice	required candidate
Location found	may disambiguate	optional candidate
Student face	not needed	remove
Guaranteed owner	unsupported promise	revise output

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **Two methods both produce the required outputs on the same five valid requests. One team claims its prototype is more accurate because it uses an app. Is that conclusion supported? Name a separate benefit the team could investigate and what it would need to record**

MAIN LESSON · SECTION 8

Practise: compare methods under the same contract

> Keep development cases separate from reserved final evaluation.

Use the same inputs, catalogue and acceptance rules

Both methods must receive the same request values and the same catalogue version, and both must be judged by the same **requirement**s. Record actual results, including errors, no-match **output**s and clarification requests.

A valid no-match result can pass; a no-match result that overlooks a **fitting** activity cannot. A guessed answer to invalid **input** fails the clarification **requirement**.

Protect the final set and document requirement changes

The cases used while designing are development cases. Keep the teacher's reserved final cases unopened until the **process** is fixed and ready for evaluation.

Do not change the **requirement**s after seeing a poor result simply to count that result as a success. If a real need changes, record the new **requirement** and rerun the relevant checks.

Run the activity baseline in W2

Use workbook W2 for the connected example "**Requirement**s and a simple **baseline**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the meaning of Quiet required No

- Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to bin requests 10 and blank

Trace requests 10 and blank. Why should they not receive the same explanation?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A useful design makes required information and promised result explicit.

Use a fresh example to explain **required input**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: An **output** label can promise more than the **process** actually knows.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **Apply the manual baseline to 12 minutes with quiet required and to 8 minutes with quiet required. Mark every catalogue row for each request, choose the required output and explain the tie rule. State what to do if the available-time field is blank instead**

Practise with less help: Run the baseline on three requests

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate two input designs on identical requests

> Supply the same catalogue and contract to both designs.

The two fictional designs

Paper two-design audit using the approved B1–B4 activity catalogue: Design A asks for available whole minutes 0–30, Quiet required Yes/No and pupil name. Design B asks only for the same minutes and quiet requirement. Both promise the first compatible catalogue activity, no match for a valid request with no fit, and clarification for invalid input. No real names or pupil data are needed.

Predict, trace and remove the unnecessary field

1. Predict whether removing the name changes any compatibility decision. Trace both designs on the same fictional cases: (12,No), (8,Yes), (blank,Yes) and (0,Yes). Show each activity's time and quiet checks, actual output and reason. Remove the unnecessary input and explain why identical correct outputs do not establish that the redesigned process is faster or easier to use. Label these paper traces, not live AI tests.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep all outputs and bounded conclusions

Case	Prediction	Observed result	What it means
Normal			
Edge			

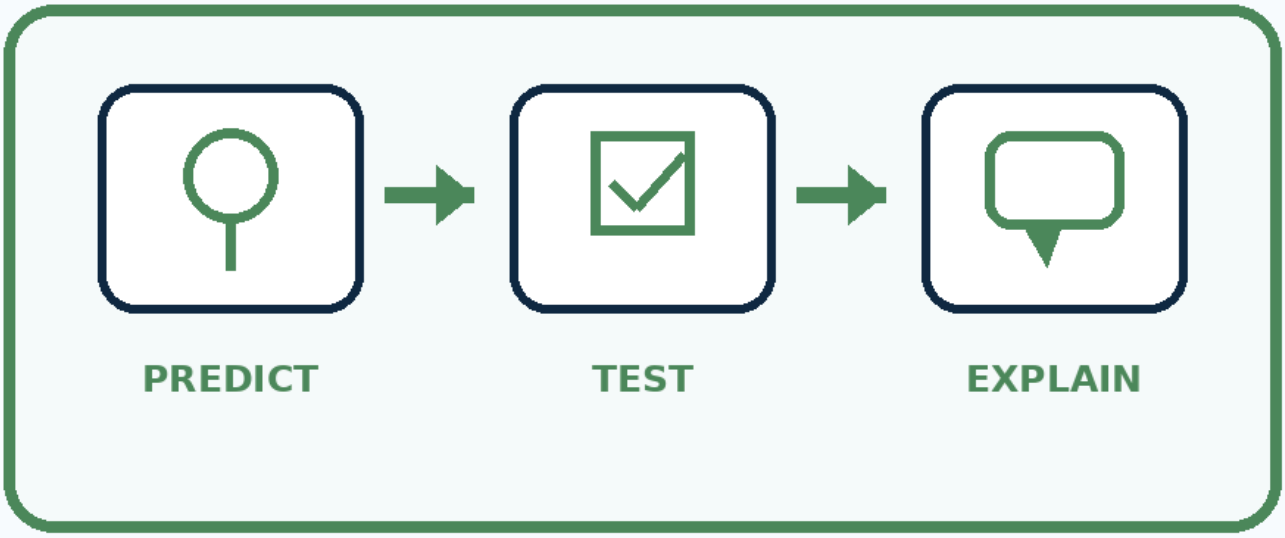
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Two-Design Input Audit

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Sketch two designs and remove an unnecessary **input**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: build the different room baseline

> Use R1/R2 and the room-capacity rule before feedback.

The independent room catalogue and requests

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a baseline for a different catalogue

My independent explanation, trace and limits

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer the contract method to the sorter portfolio

> Keep the activity project and sorter evidence distinct.

How the two contexts relate

The main final sequence continues the paper activity recommender. This section transfers its input/output-contract method to the existing sorter portfolio; it does not require a second build. Keep each catalogue and its codes labelled. The activity baseline remains the comparison method for the connected sequence.

This week’s sorter contract

THIS WEEK'S CHECKPOINT Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

Inspect fields, outputs and uncertainty

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Limit inputs and promises to the task

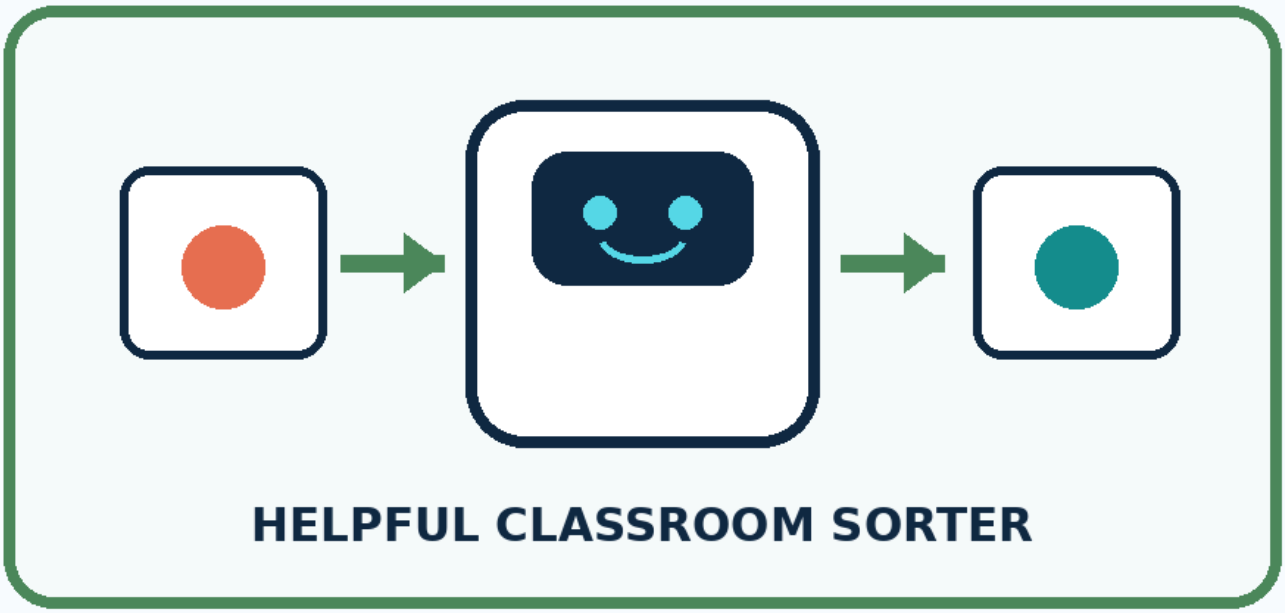
RESPONSIBLE DESIGN Every collected field creates cost and risk; every promised **output** needs **evidence**.

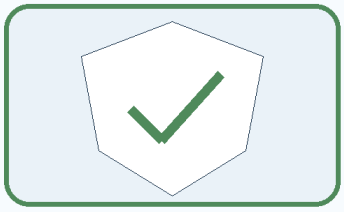
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: state the contract and its uncertainty paths

> Close your notes and distinguish clarification from no match.

Your exit explanation

- Close your notes. Explain this goal in your own words: Turn the project outcome into precise **input** and **output requirement**s

My deciding step, evidence and one remaining question

Choose a changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
required input	information necessary for the process	
optional input	information that may help but is not required	
output contract	specific result the system promises to produce	
missing-data behavior	what happens when an input is absent	

Terms for the connected explanation

Word	Meaning
Requirement	A precise condition that the solution must satisfy
Baseline	A defined starting method used for comparison
Compatible	Meeting all the stated conditions of the request

Word	Meaning
Tie rule	A stated way to choose when several options qualify
Acceptance check	A check of whether a result satisfies a requirement

Repair a claim using a word

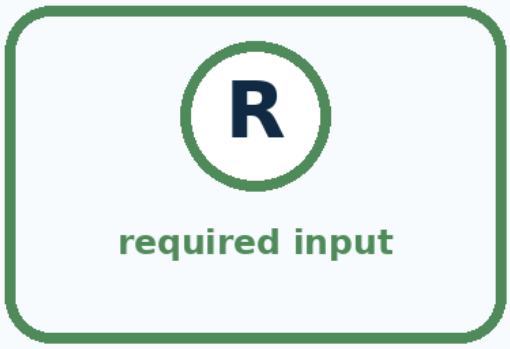
- CLAIM TO REPAIR More **input**s make every design more accurate.
- Use one vocabulary word and one piece of **evidence** to repair it:

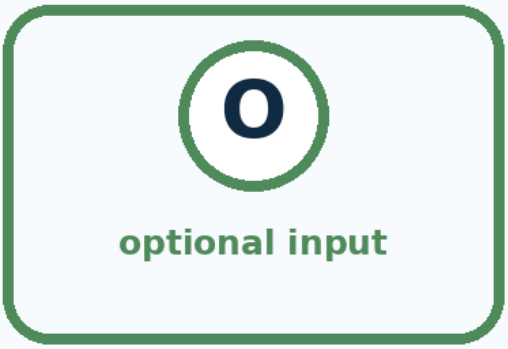
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

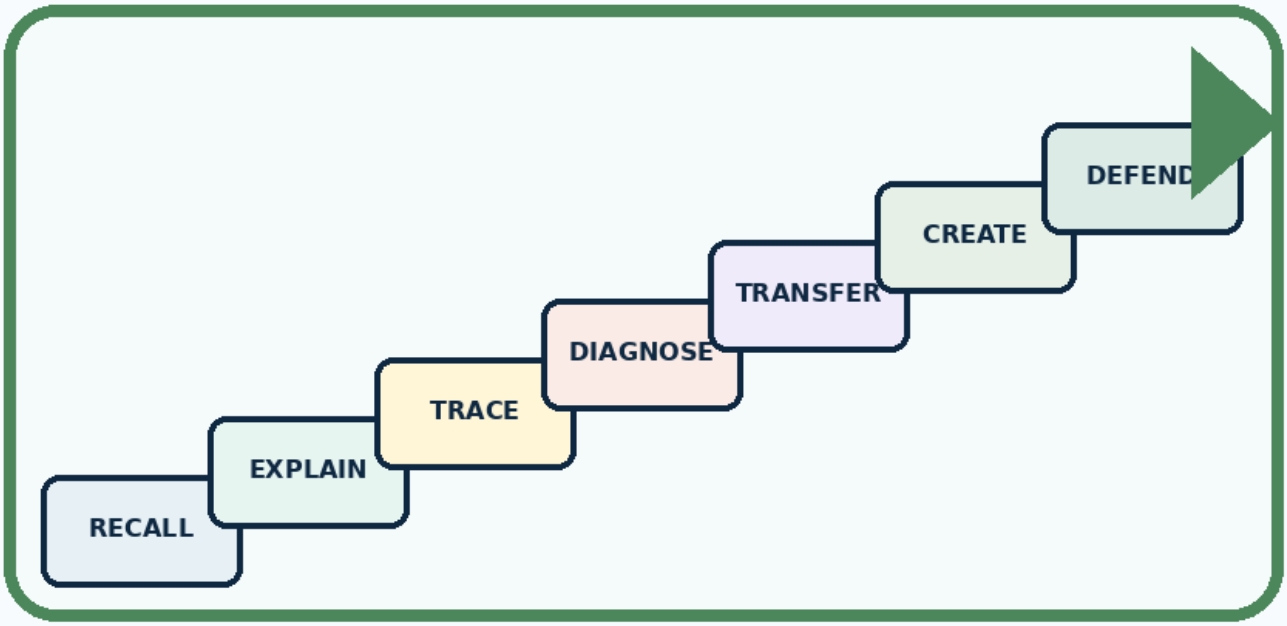
- RECALL Define **required input**.
- EXPLAIN A useful design makes required information and promised result explicit.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: More **input**s make every design more accurate.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Build a baseline for a different catalogue is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Build a baseline for a different catalogue is the selected independent case.



EXPLANATION CHECK Validate **input**s, apply every fitting condition and the stated order, and record actual results under a shared comparison contract.

Independent **evidence**:

SELF-CHECK Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

NEXT BRIDGE Carry today's **evidence** habit into: Design the Decision **Process**.

▼ **Extra practice: Design a sorter with only relevant inputs**

REFERENCE · SECTION 13

Extra practice: Design a sorter with only relevant inputs

> Try the question before opening its worked explanation.

Read the supplied case and question

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Evidence or stage	Value or result
Input	Material or a suitable image
Decision	Paper, plastic or uncertain
Output	Tray suggestion
Review	Ask a person when material is unclear

▼ Earlier wording, alternatives and picture transcripts

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Design a sorter with only relevant inputs

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Turn the project outcome into precise input and output requirements	Explain the deciding step, show a trace or test, and state one limit.
Apply a simple manual baseline to valid requests and record its results	Explain the deciding step, show a trace or test, and state one limit.
Plan a fair comparison using the same catalogue, requests and judging rules	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional classroom chooses a storage bin for model parts.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classroom chooses a storage bin for model parts.
- Which labelled step matches this detail? Report B2 and stop; save the comparison **evidence**.
- 1. **Baseline output**
- 2. Check first bin
- 3. Validate request
- 4. Check second bin
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: **Baseline output** Report B2 and stop; save the comparison **evidence**. The **baseline** is an ordered catalogue scan, not a preference ranking.

It makes expectations testable before a prototype is built. Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Changed-case question

- Changed-case practice: Change request 6 to 5. Which conclusion is supported by the supplied facts?
- 1. The **baseline** returns B1.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: The **baseline** returns B1. Its capacity equals the request, and at least includes equality.

Repair and save your work

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

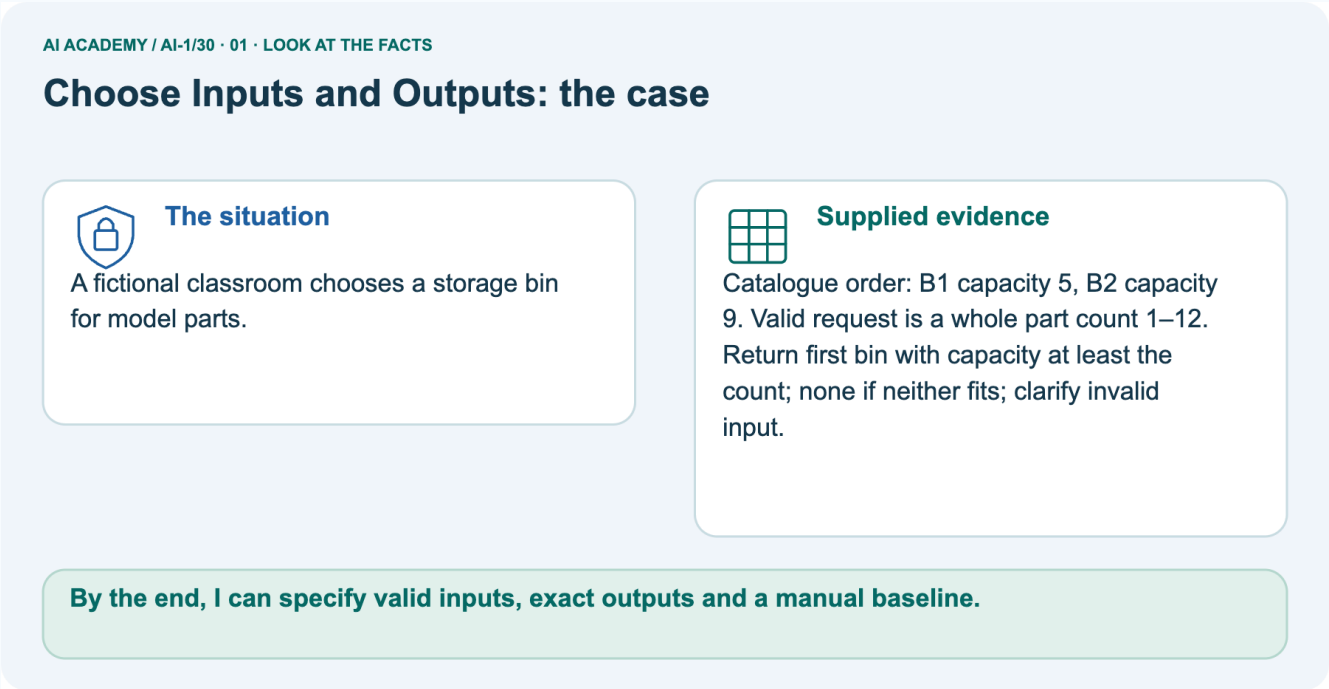
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Choose Inputs and Outputs: the case



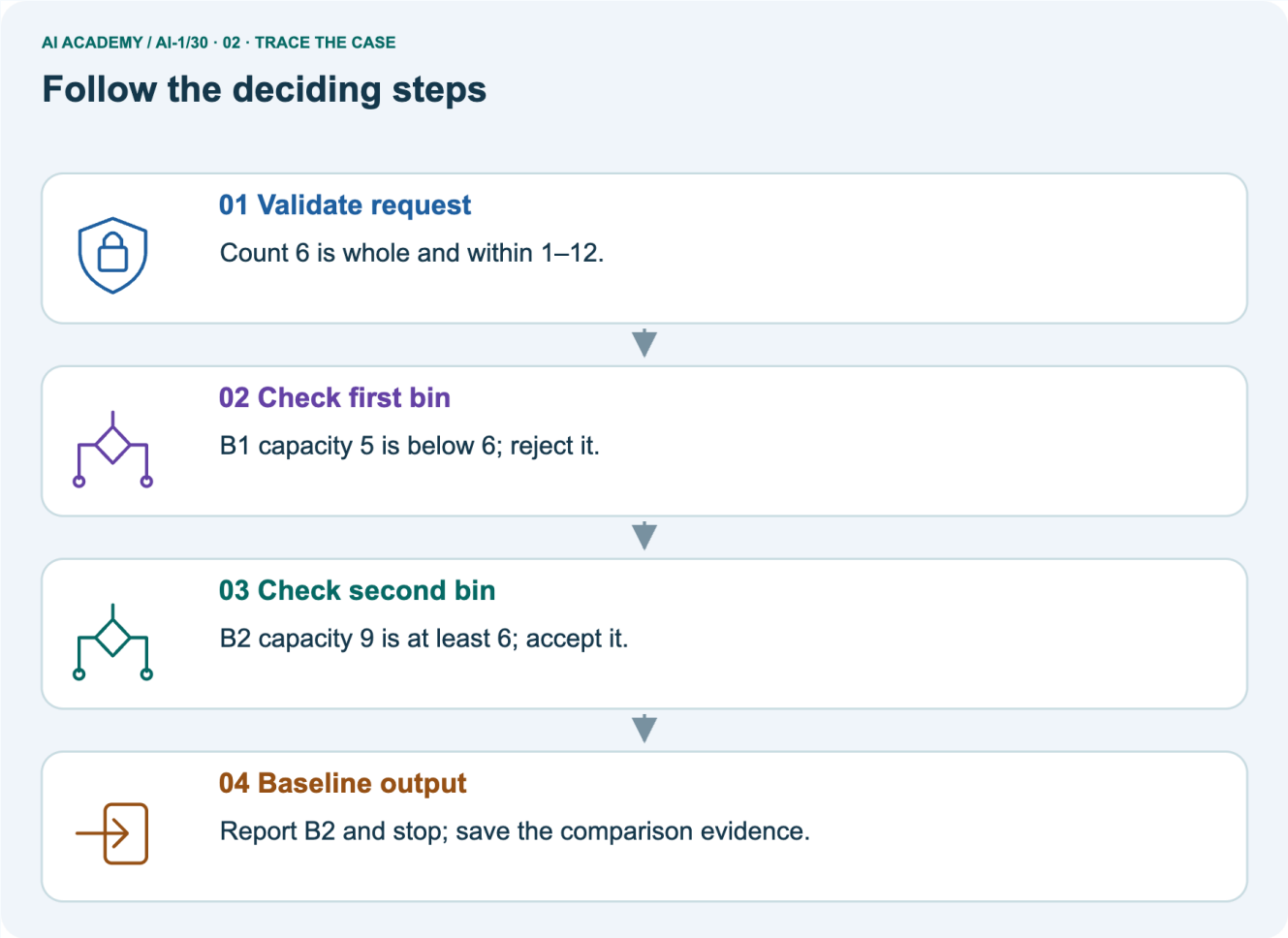
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

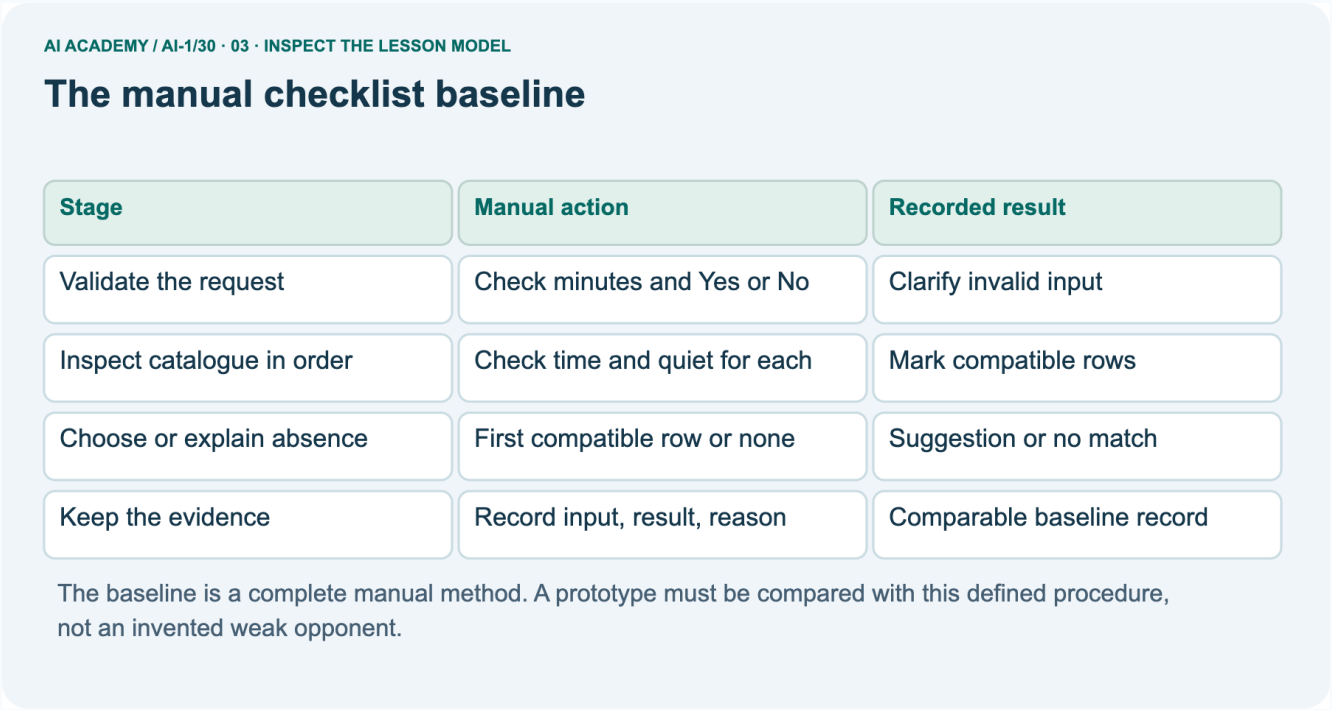


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

03. The manual checklist baseline



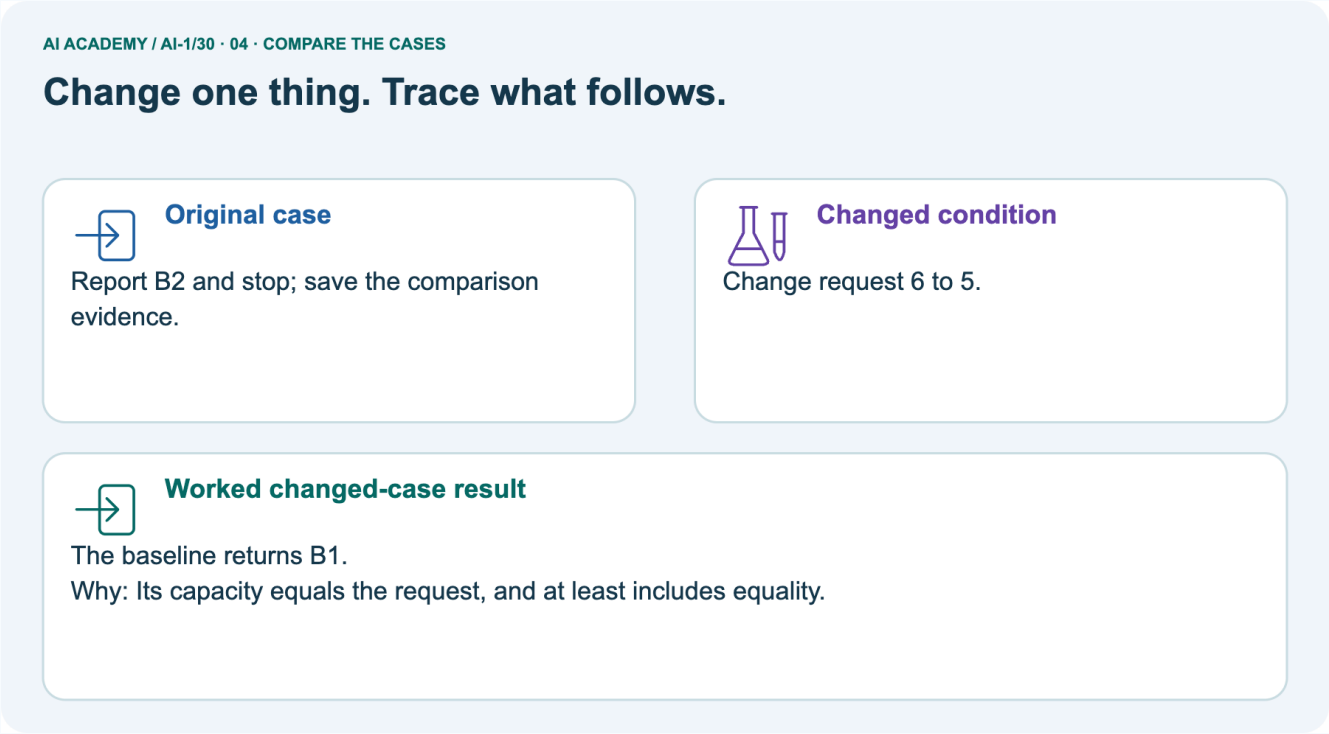
Follow the labelled picture with the matching explanation. Open the vector diagram

The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

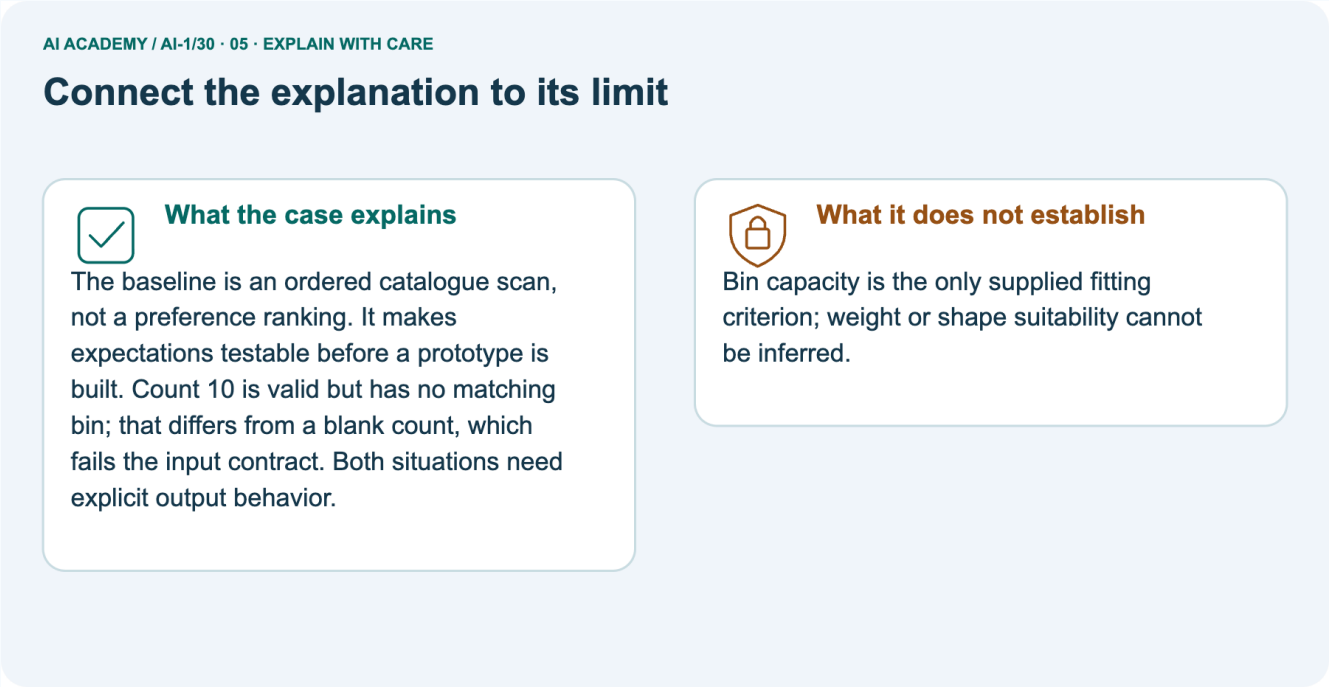


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

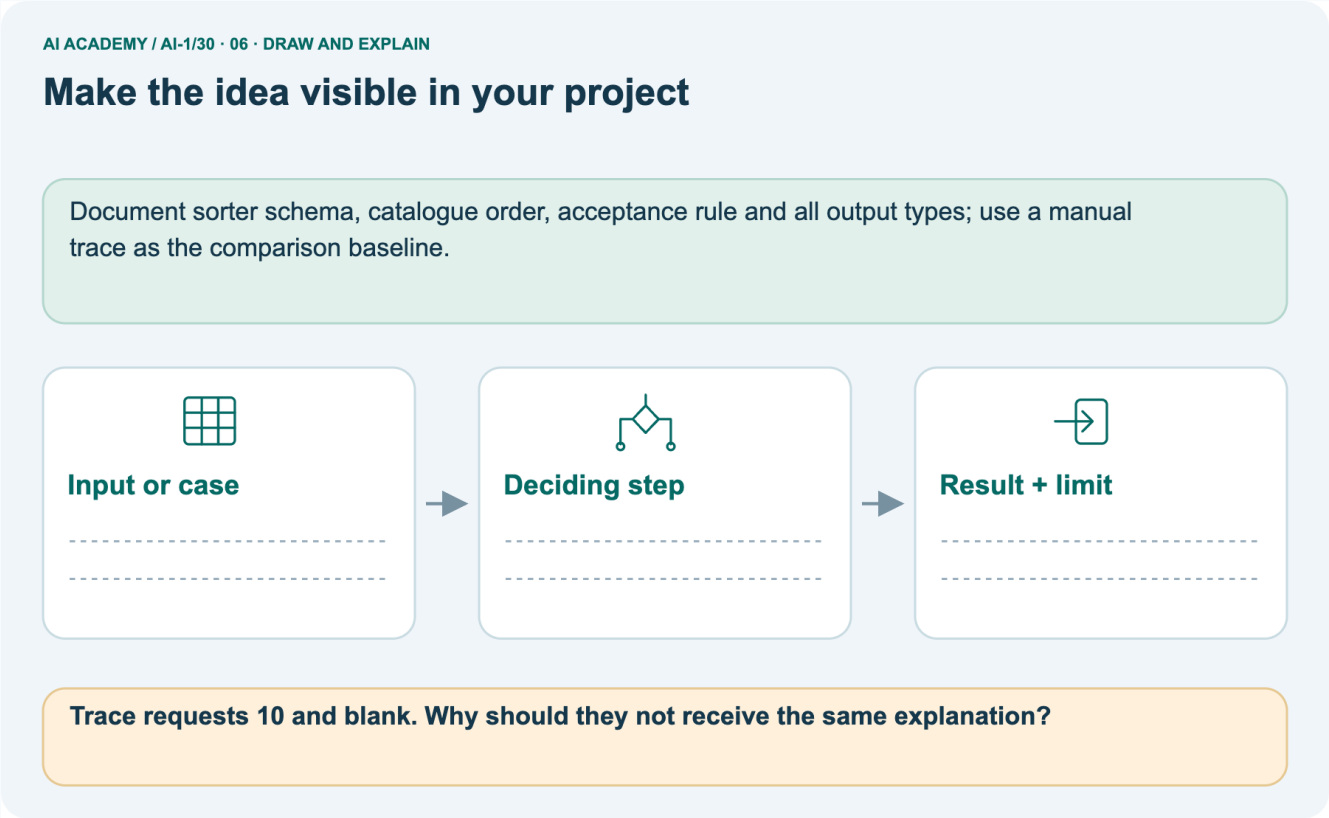
The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Choose Inputs and Outputs: the case

02. Follow the deciding steps

03. The manual checklist baseline

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.